

Extreme Programming CRC cards

CRC cards (Class-Responsibility-Collaborator cards) are a tool used in **Extreme Programming (XP)** and other **Agile software development** methodologies for **object-oriented design**. They help teams brainstorm and structure the architecture of a system by focusing on the roles and responsibilities of different classes and how they interact with one another.

Key Components of CRC Cards:

1. **Class:**
 - Represents an object or entity in the system.
 - The name of the class is written at the top of the card.
 - Example: *Product*, *User*, *Order*.
2. **Responsibilities:**
 - Defines what the class is supposed to do—its duties, behaviors, or functions.
 - Example: For a *Product* class, responsibilities could be "*calculate price with discount*" or "*update stock quantity*."
3. **Collaborators:**
 - Lists other classes that this class interacts with to fulfill its responsibilities.
 - Example: A *ShoppingCart* class might collaborate with the *Product* and *Payment* classes.

Why Use CRC Cards?

- **Simplicity:** Easy to create with just index cards or sticky notes.
- **Collaboration:** Encourages team discussions, often used in design sessions.
- **Visualization:** Makes it easy to visualize object-oriented structures without diving into code.

How CRC Cards Work in XP:

1. **Brainstorming:** The team gathers to discuss system features.
2. **Card Creation:** For each key class, a card is created listing responsibilities and collaborators.
3. **Role-Playing:** Team members "act out" interactions between classes to simulate system behavior.
4. **Refinement:** Cards are updated or reorganized as the design evolves.

Example (Sustainable Shopping Assistant App):

Class: *Product*

- **Responsibilities:**
 - Store product details (name, barcode, sustainability score)
 - Retrieve sustainability ratings
- **Collaborators:**
 - *Database* (for product data)
 - *Scanner* (to input barcode)